

Vishal Joshi

Principal Software Engineer

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(No visa sponsorship required to work in the UK)

Professional Summary

Results-driven technical leader with over 16 years of experience in architecture and development of enterprise-level applications. Expert in Java and Kotlin ecosystems with hands-on experience building scalable services and distributed data pipelines. Strong advocate of clean code practices, including SOLID and test-driven development. Proven track record of leading cross-functional technical initiatives from architecture to delivery, with expertise in building resilient, data-intensive and event-driven applications. Skilled in designing maintainable system architectures that effectively model complex business domains and workflows.

Technical Skills

- **Programming Languages:** Java 8/11/17, Kotlin, Python, SQL
- **Frameworks & Architecture:** Spring Boot, Microservices, Event-Driven Architecture
- **Messaging & Streaming:** Apache Kafka
- **Data Processing:** Apache Spark, Slurm, AWS Firehose
- **Data Systems:** MongoDB, Elasticsearch, PostgreSQL, Oracle, SOLR Index, Redis
- **Cloud:** AWS
- **Testing & Quality:** JUnit5, Spock, TDD, Pair Programming, "Test First" Methodology
- **Tools:** JMeter, Kibana, Gatling

Professional Experience

Principal Software Engineer

Tokenovate Ltd., London, UK | September 2025 – Present

- **Architecture & System Design:** Lead the architecture and system design of trade state mutation modelled against the Common Domain Model (CDM), ensuring technical rigour and consistency across complex financial workflows.
- **Engineering Leadership:** Direct an engineering team spanning Senior and Software Engineers, translating the technical vision inherited from the CTO into clear engineering direction.
- **Mentoring & Standards:** Actively mentor engineers across the team, supporting their technical growth and helping to raise overall engineering standards.
- **Product Delivery:** Working in close partnership with the Product Owner, oversee the end-to-end delivery of features with broad organisational impact, from high-level system design through to story decomposition and implementation.
- **Event-Driven Architecture:** Design and deliver event-driven architectures leveraging Apache Kafka, with services built on Spring Boot and Kotlin, ensuring systems are scalable, resilient, and aligned with enterprise standards.
- **Data Architecture:** Design solutions across both relational and non-relational data stores, ensuring scalability and resilience throughout.
- **Code Quality:** Conduct code reviews across the team, ensuring consistency, maintainability, and adherence to architectural principles throughout the codebase.
- **Design Lifecycle Ownership:** Own the full design lifecycle, from conception and stakeholder approval through to production implementation, maintaining a strong focus on engineering quality at every stage.

Project Lead, Software Engineering & AI

EMBL-EBI, Hinxton (Cambridge), UK | June 2017 – August 2025

- **Architecture & System Design:** Led the architecture and system design of new features, creating proof of concepts and presenting to leadership teams for organisation-wide roll out, ensuring scalable and resilient solutions.
- **Data Pipeline Optimisation:** Architected and optimised Apache Spark-based data processing pipelines, reducing index generation time by 83% (from 96 hours to 16 hours), demonstrating expertise in high-throughput processing of large data volumes.
- **API Design & Implementation:** Developed RESTful web services and APIs for complex biological data, including the first Large Language Model (LLM) API at EBI that increased literature curation speed by 400%.
- **Automated Annotation (ARBA):** Developed ARBA, a data pipeline powering an LLM copilot for protein curation, improving curation speed by 400%.
- **Industry Collaboration (ProtNLM):** Collaborated with Google DeepMind on the development of ProtNLM, contributing to protein function prediction using natural language models.
- **Technical Leadership:** Mentored engineering teams through code reviews, pair programming, and technical sessions, fostering a collaborative environment that emphasises code quality and best practices.
- **Performance Engineering:** Led a major refactor of prediction pipelines, enabling integration of 49 million additional high-throughput predictions by optimising memory, storage, and execution patterns.
- **Cloud Architecture:** Designed and deployed a cluster of RESTful web services on Google Cloud Platform (GCP), improving range query performance from 5000ms to 100ms through sophisticated data modelling and optimisation.
- **CI/CD Implementation:** Spearheaded the implementation of the first CI/CD pipeline, transitioning from manual releases to an efficient, automated system using GitLab and DevOps practices, supporting 20 large data pipelines.
- **Production System Reliability:** Responsible for investigating, diagnosing, and providing resolutions to production issues in large-scale data pipelines, web services, and applications, ensuring system resilience and reliability.

Applications Engineer

Rakuten Inc., Tokyo, Japan | June 2016 – May 2017

- **API Development:** Developed intelligent Search API and History API that personalised results based on user profiling and provided real-time booking data, increasing transaction conversions by 18%.
- **Microservices Architecture:** Implemented RESTful APIs using microservices architecture, improving system scalability and maintainability through modular design principles.
- **Test-Driven Development:** Created the PATW API using TDD and BDD methodologies, increasing partner business transactions from 45 million JPY to 54.9 million JPY during peak season.
- **Performance Engineering:** Designed and executed performance testing scenarios using JMeter and Gatling, simulating varying traffic conditions to ensure platform stability under high loads.

Software Engineer

Equal Experts, Pune, India | June 2014 – May 2016

- **Microservices Implementation:** Developed a suite of microservices (Profile API, Order API, and Notifications API) for O2 Priority UK, each designed with distinct scalability needs and communication patterns.
- **Message-Based Integration:** Integrated notification services with multiple messaging platforms including Google Cloud Messaging and Apple Push Notification Service, implementing efficient message handling patterns.
- **Requirements Analysis:** Actively translated business requirements into technical specifications, creating detailed system design documents and architectural decision records (ADRs).
- **Containerisation & Deployment:** Implemented Docker-based containerisation for microservices to facilitate managed deployments with rolling builds using Ansible for continuous deployment.
- **Resilient System Design:** Delivered technical presentations on resilient software design principles and circuit breaker patterns using Hystrix, demonstrating expertise in building fault-tolerant systems.

System Analyst

Cybage Softwares Pvt. Ltd., Pune, India | October 2010 – June 2014

- **Technical Leadership:** Designed, developed, and led a team of engineers for TX-SMART, implementing an offline queuing mechanism for fault-tolerant event synchronisation.
- **Financial Systems Integration:** Developed and maintained software components for settling credit card, debit card, and bank account transactions with SWIFT gateway systems, gaining exposure to financial transaction processing.
- **Web Application Architecture:** Led development of critical map, SMS, and voice features for the Heimdall project in a single-page web application built with Ember JS.

Education

Bachelor of Technology (Computer Sciences)

Punjab Technical University | May 2006 – May 2010

- Distinction holder throughout 4 years
- Developed compiler components during institutional training

Certifications

- Machine Learning by Stanford University – Coursera, June 2020
- ChatGPT Prompt Engineering for Developers – Coursera, 2024
- Building Systems with ChatGPT API – Coursera, 2024

Achievements & Recognition

- Delivered technical talks on resilient software design at international conferences, including ISMB 2023 and 2024
- Received commendations for best practices in code development at EMBL-EBI
- Achieved three promotions from Software Engineer to System Analyst at Cybage within three years
- Received multiple client appreciations at Cybage and Equal Experts for technical excellence